

This isn't the first initiative to promote RTS power systems. Back in 2014, the government introduced the Rooftop Solar Programme, aiming for a total installed capacity of 40 GW by 2022. However, the goal wasn't met, prompting the government to extend the deadline to 2026. The new scheme is another attempt to achieve the 40 GW rooftop solar capacity target. "Since 2016, household RTS capacity in India has grown to 2.7 GW with 700,000-800,000 households being powered by solar energy. It can take another five years to set up an additional 25-30 GW," says Vikas Agarwal, Co-Founder and Chief Business Officer of Credit Fair, a consumer lending fintech start-up.

According to the Ministry of New and Renewable Energy's (MNRE) website, the total installed capacity of solar power in India has reached around 82 GW as of April 2024. Meanwhile, grid-connected RTS is around 12.01 GW as of April 2024. "Even though the MNRE launched the Rooftop Solar Programme Phase-II in 2019 and the PM Surya Ghar [Muft Bijli] Yojana, the adoption of RTS has not met expectations due to a lack of awareness and financing options. However, with government support and initiatives like no-cost EMIs, we are hopeful of increased demand and adoption of RTS," adds Agarwal.

Here is a guide for those who want to instal a residential RTS power system.

COUNT THE COST

Though installing rooftop solar panels is a bright idea, there is a substantial cost involved as solar panels are mounted on building roofs and are connected to the main power supply unit. "The set-up reduces dependence on grid-connected electricity, resulting in significant cost savings for consumers," says Sandeep Sahoo, Partner at Odisha-based Prakruti

Energy, a wholesaler of solar panels. However, there is an initial fixed cost involved. "For example, for a 1 kW connection, the set-up costs around ₹90,000, for 2 kW the expenditure is around ₹1.5 lakh, and for 3 kW, the cost is ₹2 lakh," he adds.

Apart from the sanctioned load, the cost of setting up also depends on the type of solar panels. For instance, the primary types are poly panels, mono panels, and bifacial solar panels.

Polycrystalline or poly panels feature multiple crystals in each cell, giving them a blue look. They are made by melting raw silicon, which results in several crystals forming in each cell. Therefore, they have a lower efficiency compared to monocrystalline panels as the presence of many crystals can inhibit the movement of electrons. This reduces the panel's energy output.

However, they offer a more affordable solution and are ideal for large roofs without any space constraints. Mono panels are recognised by their dark black colour and uniform appearance. They are built with a single silicon wafer and are more expensive than poly panels. Lastly, bifacial solar panels can absorb sunlight from both sides. They are designed to generate power from the front side while capturing reflected sunlight from the rear side. This dual action can significantly increase the total energy yield.

"Each type of solar rooftop panel offers unique advantages. The choice ultimately depends on your budget, space constraints, amount of sunlight received, and electricity needs," says Sahoo.

He advises customers to opt for mono panels as the government subsidy calculation is based on this type of panels.

Some of the popular names in the rooftop solar panels space are Tata Power, Adani Solar, Jakson Power, Waaree Energies, Vikram Solar, RenewSys and Servotech

MIND IT



To be eligible for subsidy, installation should not exceed 85% of the sanctioned load



Subsidy is applicable only for panels made in India



Current electricity connection should be in your name



Ensure consistency in bank account, electricity bill, and Aadhaar card names



Without grid power, one can't use solar energy. An inverter is needed as a backup when there is no power from the grid



Installation requires shadow-free space. For example, installing a 3 kW solar plant requires approximately 300 sq. ft of shadow-free space